
Where Does Platonic Convergence Live? Pointwise Curvature of a Galaxy Embedding Manifold and What a Linear Map Can See

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Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations on galaxy images
3 using Mean-KNN (MKNN) and Centered Kernal Analysis (CKA). We ask whether
4 that convergence is concentrated where the embedding manifold is flat. A point-
5 wise curvature field, obtained by differentiating a trained decoder and validated
6 against analytic truth in that regime, finds mean-curvature magnitude correlating
7 negatively with cross-survey neighbour agreement at every latent dimension, with
8 local density explaining 50 to 80 percent of it. A second-order expansion of a
9 linear map restricted to the manifold explains why: a readout sees the second fun-
10 damental form contracted with its normal component, $\langle w_N, \Pi \rangle$, and an alignment
11 sees the relative second fundamental form of the two embeddings, neither visi-
12 ble to mean-curvature magnitude at large codimension. On a unit-normalized em-
13 bedding that Hessian splits into a sphere term, proportional to the readout’s own
14 value, and a shape term. On a synthetic surface at production scale the sphere term
15 reads -0.74 to -0.80 across samplings that swing the $\|H\|$ partial from -0.29 to
16 $+0.26$; on 86,471 labelled galaxies it is negative on every label, the mismatch be-
17 tween the label’s own Hessian and the shape term is negative wherever the label
18 has local structure, and the shape term alone takes the label-dependent sign that
19 mismatch predicts. Convergence tests must condition on relative curvature, not on
20 one model’s.

21 1 Introduction

22 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
23 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on the sky:
24 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) and embedded by
25 a family of different vision models share 0.4 to 2 percent of their nearest neighbours. We ask where
26 on the manifold that agreement lives: if two models disagree most where the embedding surface
27 bends sharply, local mean curvature should associate negatively with local cross-survey agreement.
28 We test this claim using a pointwise mean-curvature estimation from a trained autoencoder on the
29 embedding space.

30 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
31 trained decoder, validated against analytic truth in the production regime; (ii) a density-controlled
32 test of the curvature–alignment prediction under two probes, which finds no effect free of density;
33 (iii) the residual expansion of a linear map on a submanifold, which names the curvature such a map
34 responds to and shows mean-curvature magnitude is blind to it; and (iv) a known-surface and a real-

35 data test on the linear-readout side, where the expansion’s terms behave as predicted and account for
 36 a label-dependent sign.

37 2 Pointwise curvature from a decoder

38 **Data.** DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both
 39 HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019], from the Platonic Universe re-
 40 lease [Duraphe et al., 2025], from which we draw 10,000 rows once with a recorded seed. A second
 41 set of 86,471 single-survey ViT-B embeddings with catalogue labels (r -band magnitude, photometric
 42 redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the read-
 43 out experiment. Embeddings are unit-normalized (data on S^{767}). We then estimate the manifold’s
 44 first homology ($\beta_1 = 0$ under bootstrap bands [Fasy et al., 2014]), thus we one Euclidean chart is
 45 appropriate for the autoencoder.

46 **Instrument.** Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden lay-
 47 ers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With
 48 $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form
 49 $\Pi = P_N D^2 F(z)$, the mean-curvature vector is $H = \text{tr}_g \Pi$, by automatic differentiation checked
 50 against finite differences. On the unit sphere every d -submanifold carries a radial component of H
 51 equal to $-d$ that says nothing about shape. To account for this, the map differentiated is $F/\|F\|$
 52 (after which $H_{\text{rad}} = -d$ to 10^{-14}) and $\|H_{\text{tan}}\|$ carries the shape. Decoder curvature has served as
 53 a regularizer [Lee and Park, 2023] and been read from templates [Acosta et al., 2022]; here it is an
 54 instrument, validated against analytic fixtures.

55 **Validation.** Scored on surfaces with closed-form curvature on four axes: rank (Spearman against
 56 true $\|H\|$), direction (cosine of H vectors), magnitude ratio and calibration. At $D = 768$ an in-sphere
 57 surface $G(z) = Q \text{ normalize}([\phi(z); 0.8 h(z); 0 \dots])$ (ϕ the inverse stereographic map $\mathbb{R}^{16} \rightarrow S^{16}$,
 58 h four Gaussian bumps, Q a fixed rotation, span 18 dimensions) at $n = 86,471$, $k = 2,048$ gives
 59 rank 0.94, direction cosine 0.999, magnitude ratio 1.00, and 0.75 rank under noise at 25% of patch
 60 radius.

61 3 Curvature and cross-survey convergence

62 **Probe.** Per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k =$
 63 20 between the HSC and Legacy Survey embeddings, the Platonic Universe statistic made local,
 64 correlated with $\|H\|$ over all 10,000 points at $d \in \{20, 25, 32\}$. Centered kernel alignment [Kornblith
 65 et al., 2019] is the second probe.

66 **Result.** The raw association $\rho(\|H\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$
 67 ($p < 10^{-5}$, $< 10^{-5}$, 0.009), in the predicted direction. But k -NN statistics are density-
 68 sensitive [Radovanović et al., 2010]: density correlates with agreement at -0.21 at every d and with
 69 $\|H\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 , -0.066 and -0.022 :
 70 78%, 49% and 8% of the association removed. Under a permutation null within density strata only
 71 $d = 25$ survives ($p < 5 \times 10^{-5}$; replicating across three decoder seeds; the CKA tertile gap within
 72 strata agrees at $d = 20$ and 25 , -0.05 to -0.08 against a null band of ± 0.017). Replacing $\|H\|$ by
 73 $\|H_{\text{tan}}\|$ collapses the $d = 25$ partial to -0.023 , into the range of the two d that fail. Convergence is
 74 not detectably concentrated by mean-curvature magnitude once density is controlled.

75 4 What a linear map sees on a curved manifold

76 Let $M \subset \mathbb{R}^d$ be d -dimensional; at x_0 take an orthonormal tangent frame J , the second fundamental
 77 form Π (normal-valued, symmetric $d \times d$), $H = \text{tr } \Pi$, and for a vector w its tangent and normal parts
 78 $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$, so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

79 To second order, a linear function is as non-linear on M as M bends in that function’s normal direc-
 80 tion [Lee, 2018]. For $M \subset S^{D-1}$ the normal bundle splits into the sphere’s radial direction and the

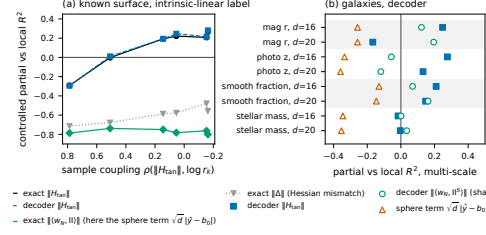


Figure 1: (a) Known surface, intrinsically linear label, exact geometry: three-control partial against local R^2 as the sample’s density–curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$: partial under the multi-scale density control for the decoder’s $\|H_{\tan}\|$ (filled squares), the shape term (open circles) and the sphere term (open triangles).

rest, $\Pi = \Pi^S - g \hat{x}$, so $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x}) g$ (2): a linear readout of a unit-normalized embedding is bent by the manifold’s shape and, whatever the manifold, by the sphere in proportion to the readout’s own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$ in normal coordinates (the $\frac{1}{2}$ below uses Gaussian fourth moments), with $\bar{c}$ the patch-mean residual, $\delta = \nabla_M y - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$,

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

For two embeddings F, G of the same objects in a common chart u and a linear map A with $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G - A \Pi_F](u, u) + O(|u|^3)$, with Γ the induced connections: the second-order obstruction to linear alignment is a connection mismatch plus the *relative* second fundamental form, and reduces to the latter when the induced metrics agree; its normal part is $\Pi_G - P_N^G A \Pi_F$.

So the curvature a linear map pays for is $\langle w_N, \Pi \rangle$ or $\Pi_G - P_N^G A \Pi_F$; $\|H\|$ bounds only a trace and need not correlate with either at codimension 750; the term carries s^4 , so patch radius, hence density, multiplies it; and when the label has intrinsic curvature of its own, Δ carries the residual and the curvature piece can take either sign.

5 The probe-facing curvature, on a known surface and on galaxies

The readout side can be tested directly: w is known and Π comes from the decoder. Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors’ 2,048-neighbourhoods; rank-partial Spearman against local R^2 (Freedman–Lane permutation p -values [Freedman and Lane, 1983]) controlling for log patch radius, local label variance and count and, as a stronger control, log radius at $k \in \{16, 64, 256, 1024, 2048\}$. Per anchor we compute $\|H_{\tan}\|$ and the probe-facing Hessian $\|\langle w_N, \Pi \rangle\|_g$ with $w_N = w - Jg^{-1}J^\top w$, split by Eq. (2) into its shape and sphere terms.

Known surface. The validation surface is resampled with weight $\|H_{\tan}\|^\gamma$, so only where the points sit changes (Figure 1a). The $\|H_{\tan}\|$ partial follows the sampling from -0.29 at coupling $+0.80$ through zero to a plateau near $+0.22$, and the decoder reproduces it. The probe-facing Hessian $\|\langle w_N, \Pi \rangle\|$ reads -0.74 to -0.80 at every sampling. On this surface it is the sphere term of Eq. (2): rank 0.996–1.000 with $\sqrt{d}|\hat{y} - b_0|$, the in-sphere part 20 to 40 times smaller (one in-sphere normal direction) and, where resolvable, following $\|H_{\tan}\|$. The fixture thus confirms the expansion’s sphere term: an intrinsically linear label loses local R^2 where the readout’s own value is large, whatever the sampling. With a nonlinear label the sphere term reads null and $\|\Delta\|$ carries the partial (-0.2 to -0.6), as Eq. (3) predicts.

Galaxies. Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99 with the full norm). To read it through Eq. (3) we estimate $\text{Hess}_M y$ per anchor by a least-squares quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\top(x - x_0)$ of its 2,048 neighbours, where the quadratic coefficient is the covariant Hessian. Three regularities (Table 1). The sphere term is negative on every label and both d (-0.13 to

label	$\ H_{\text{tan}}\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control; alignment is $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$. * not significant at 0.05; the three-control partials agree in sign in every significant cell.

−0.36), as on the fixture. Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar mass is null throughout) the mismatch is negative (−0.09 to −0.45) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive (+0.07 to +0.43): local R^2 is higher where the manifold bends in the readout’s direction the way the label bends. The shape term alone is therefore label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift) because it is one of three terms in $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, \langle w_N, \Pi \rangle \rangle + \|\langle w_N, \Pi \rangle\|^2$: its sign follows the cross term’s where that is nonzero and the remaining square where it is null. $\|H_{\text{tan}}\|$ changes sign between $d = 16$ and 20 for magnitude: it sits on the curve of Figure 1a (coupling to radius −0.60). The same regression applied to the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median metric cosine with the decoder’s 0.55–0.65), and an independent split-half quadratic-chart estimator of Π gives the same shape-term sign on three of four labels.

Across encoders. The quadratic-chart estimator’s mean-bending trace, run on five encoders at the same 512 anchors ($d = 16$, $k = 2,048$: ViT-B, DINOv3, CLIP-B, ConvNeXt-B, ViT-L), gives density-controlled associations with the magnitude probe’s local R^2 of −0.24, 0.00, +0.27, +0.23 and −0.19. A trace statistic has no encoder-stable sign, as Eq. (1) predicts for a quantity the readout does not see; the terms of Eq. (3) are what a cross-encoder test should carry.

6 Discussion

The design of Section 3 could not have found curvature-concentrated convergence: MKNN and CKA are invariant to rigid motions of either embedding, so they cannot depend on one embedding’s curvature alone; a per-point convergence score can be conditioned on curvature only through a relative quantity such as $\Pi_G - P_N^G A \Pi_F$, which needs both embeddings’ second fundamental forms and a map between tangent frames. A pilot at $d = 20$ (2,048 anchors) finds a global ridge map carries HSC tangent frames into the Legacy embedding with 44% residual and no density-free association of $\|\Pi_G(L, L) - P_N^G A \Pi_F\|$ with MKNN (+0.04, n.s.; a locally fitted tangent map gives −0.18 to −0.24, a decoder-free estimate +0.13): the obstruction is already first order. On the readout side, where the map is known, the sphere term and the Hessian mismatch carry the predicted negative sign on the known surface and on galaxies, and the shape term is label-signed because it enters through its alignment with the label’s own Hessian.

Limitations. One survey pair; the label’s Hessian is a noisy local estimate (R^2 gain over linear 0.10–0.17; a 136-coefficient fit of the same kind failed a split-half reliability gate); the sphere term is not separated from ridge shrinkage of extreme predictions; permutation p -values assume anchor exchangeability under overlapping patches.

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